

Antiderivatives and Riemann Sums

ANSWER KEY



Basic antiderivatives

- 1 Find the general antiderivative of each function:

a $f(x) = x^4 = \frac{x^5}{5} + C$

b $f(x) = \frac{1}{x} = \ln|x| + C$

c $f(x) = e^x = e^x + C$

d $f(x) = \sec^2 x = \tan x + C$

- 2 Find $\int (3x^2 - 4x + 5) dx$.

$$= x^3 - 2x^2 + 5x + C.$$

- 3 Find the particular antiderivative $F(x)$ of $f(x) = 6x^2 - 4$ satisfying $F(1) = 5$.

$$F(x) = 2x^3 - 4x + C. F(1) = 2 - 4 + C = 5 \Rightarrow C = 7. F(x) = 2x^3 - 4x + 7.$$

Riemann sums: left, right, and midpoint

- 4 Approximate $\int_0^4 x^2 dx$ using a Left Riemann Sum with $n = 4$ equal subintervals.

$$\Delta x = 1; \text{ left endpoints } x = 0, 1, 2, 3: L_4 = 1[0^2 + 1^2 + 2^2 + 3^2] = 1[0 + 1 + 4 + 9] = 14.$$

- 5 Approximate $\int_0^4 x^2 dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.

$$\Delta x = 1; \text{ right endpoints } x = 1, 2, 3, 4: R_4 = 1[1 + 4 + 9 + 16] = 30.$$

- 6 Approximate $\int_0^4 x^2 dx$ using a Midpoint Riemann Sum with $n = 4$ equal subintervals.

$$\Delta x = 1; \text{ midpoints } x = 0.5, 1.5, 2.5, 3.5: M_4 = 1[0.25 + 2.25 + 6.25 + 12.25] = 21. \text{ (True value: } \frac{64}{3} \approx 21.33.)$$

The Trapezoidal Rule

- 7 Use the Trapezoidal Rule with $n = 4$ to approximate $\int_0^4 x^2 dx$.

$$T_4 = \frac{L_4 + R_4}{2} = \frac{14 + 30}{2} = 22.$$

- 8 A table gives $f(0) = 2$, $f(2) = 5$, $f(4) = 9$, $f(6) = 8$. Use the Trapezoidal Rule with these unequal-width data (widths of 2 each) to approximate $\int_0^6 f(x) dx$.

With 3 equal subintervals of width 2:

$$T = 2 \left[\frac{f(0) + f(2)}{2} + \frac{f(2) + f(4)}{2} + \frac{f(4) + f(6)}{2} \right] = 2 \left[\frac{7}{2} + \frac{14}{2} + \frac{17}{2} \right] = 7 + 14 + 17 = 38.$$

Sums as limits: the definite integral

- 9 Explain how the definite integral $\int_a^b f(x) dx$ is defined as a limit of Riemann sums, and write the limit expression using sigma notation.

$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$, where $\Delta x = \frac{b-a}{n}$ and x_i^* is any sample point in the i -th subinterval — as $n \rightarrow \infty$ (subintervals shrink to zero width), all Riemann sum types (left, right, midpoint) converge to the same value, the exact signed area under f .

- 10 Determine whether the Left Riemann Sum overestimates or underestimates $\int_0^4 x^2 dx$, and explain why using the shape of the graph.

Since $f(x) = x^2$ is increasing on $[0, 4]$, the left endpoint of each subinterval gives the smallest function value on that subinterval, so the Left Riemann Sum UNDERESTIMATES the true area (confirmed: $L_4 = 14 < \frac{64}{3} \approx 21.33$).

Riemann sums from a table of values

- 11 A table gives $f(0) = 3$, $f(1) = 5$, $f(2) = 8$, $f(3) = 10$, $f(4) = 9$. Use a Right Riemann Sum with 4 subintervals of width 1 to approximate $\int_0^4 f(x) dx$.

$$R_4 = 1[f(1) + f(2) + f(3) + f(4)] = 5 + 8 + 10 + 9 = 32.$$

Antiderivatives of trigonometric and exponential functions

- 12 Find $\int (3\sin x + 2e^x) dx$.

$$= -3\cos x + 2e^x + C.$$

- 13 Find the particular antiderivative $F(x)$ of $f(x) = \sec^2 x$ satisfying $F(0) = 2$.

$$F(x) = \tan x + C. F(0) = 0 + C = 2 \Rightarrow C = 2. F(x) = \tan x + 2.$$

Visualizing a Riemann sum approximation

- 14 The graph shows $f(x) = x^2$ on $[0, 4]$. Use the graph to explain why the Right Riemann Sum $R_4 = 30$ (found earlier) overestimates the true area, and why the Left Riemann Sum $L_4 = 14$ underestimates it.

Since $f(x) = x^2$ is increasing on $[0, 4]$, right-endpoint rectangles reach up to the curve's height at the RIGHT edge of each strip, so their tops poke above the curve, overestimating the area. Left-endpoint rectangles use the height at the LEFT edge (lower, since f is increasing), so their tops stay below the curve, underestimating the area.

An applied antiderivative problem

- 15 A factory's production rate is $r(t) = 100 + 20t$ units/day, where t is days since a process change. Find the total production function $P(t)$ given $P(0) = 0$, then find the total production over the first 10 days.

$$P(t) = \int r(t) dt = 100t + 10t^2 + C. \text{ Using } P(0) = 0: C = 0. P(t) = 100t + 10t^2.$$
$$P(10) = 1000 + 1000 = 2000 \text{ units.}$$

More antiderivative practice

- 16 Find the general antiderivative of $f(x) = 4x^3 - \frac{2}{x^2} + \cos x$.

$$= x^4 + \frac{2}{x} + \sin x + C.$$

- 17 Find the particular antiderivative $F(x)$ of $f(x) = x^2 - 3$ satisfying $F(0) = 4$.

$$F(x) = \frac{x^3}{3} - 3x + C. F(0) = 0 - 0 + C = 4 \Rightarrow C = 4. F(x) = \frac{x^3}{3} - 3x + 4.$$

Comparing Riemann sums to the exact integral

- 18 For $\int_0^4 x^2 dx$, the exact value is $\frac{64}{3} \approx 21.33$. Rank the four approximations found earlier — $L_4 = 14$, $R_4 = 30$, $M_4 = 21$, $T_4 = 22$ — from closest to farthest from the exact value, and explain which type of approximation (midpoint or trapezoidal) tends to be more accurate for a smooth curving function.

Distances from 21.33: M_4 ($|21 - 21.33| = 0.33$), T_4 ($|22 - 21.33| = 0.67$), L_4 ($|14 - 21.33| = 7.33$), R_4 ($|30 - 21.33| = 8.67$). Order (closest to farthest): M_4, T_4, L_4, R_4 . The midpoint approximation tends to be more accurate than the trapezoidal for smooth curves, because its errors are typically about half the size and opposite in sign to the trapezoidal error.

Multiple choice — easy 15

- 19 Find C in the particular antiderivative $F(x) = x^3 - 7x + C$ of $f(x) = 3x^2 - 7$, satisfying $F(-1) = -8$.

- a. -8
- b. -12
- c. -14 ✓
- d. -16

$$F(-1) = (-1)^3 - 7(-1) + C = 6 + C = -8, \text{ so } C = -14.$$

20 Find C in the particular antiderivative $F(x) = x^3 + 3x + C$ of $f(x) = 3x^2 + 3$, satisfying $F(3) = -5$.

- a. -39
- b. -43
- c. -41 ✓
- d. -5

$$F(3) = (3)^3 + 3(3) + C = 36 + C = -5, \text{ so } C = -41.$$

21 Find C in the particular antiderivative $F(x) = -2x^3 - 5x + C$ of $f(x) = -6x^2 - 5$, satisfying $F(2) = -9$.

- a. 17 ✓
- b. -9
- c. 15
- d. 19

$$F(2) = -2(2)^3 - 5(2) + C = -26 + C = -9, \text{ so } C = 17.$$

22 Find C in the particular antiderivative $F(x) = x^3 + 7x + C$ of $f(x) = 3x^2 + 7$, satisfying $F(-3) = -2$.

- a. 46 ✓
- b. 48
- c. 44
- d. -2

$$F(-3) = (-3)^3 + 7(-3) + C = -48 + C = -2, \text{ so } C = 46.$$

23 Find C in the particular antiderivative $F(x) = 2x^3 + 9x + C$ of $f(x) = 6x^2 + 9$, satisfying $F(0) = 9$.

- a. 7
- b. 9 ✓
- c. 10
- d. 11

$$F(0) = 2(0)^3 + 9(0) + C = 0 + C = 9, \text{ so } C = 9.$$

24 Find C in the particular antiderivative $F(x) = 2x^3 + 3x + C$ of $f(x) = 6x^2 + 3$, satisfying $F(-3) = 6$.

a. 6

b. 69 ✓

c. 71

d. 67

$$F(-3) = 2(-3)^3 + 3(-3) + C = -63 + C = 6, \text{ so } C = 69.$$

25 Find C in the particular antiderivative $F(x) = 2x^3 + x + C$ of $f(x) = 6x^2 + 1$, satisfying $F(3) = -8$.

a. -65 ✓

b. -8

c. -67

d. -63

$$F(3) = 2(3)^3 + (3) + C = 57 + C = -8, \text{ so } C = -65.$$

26 Find C in the particular antiderivative $F(x) = -x^3 - 4x + C$ of $f(x) = -3x^2 - 4$, satisfying $F(-3) = 8$.

a. 8

b. -29

c. -33

d. -31 ✓

$$F(-3) = -(-3)^3 - 4(-3) + C = 39 + C = 8, \text{ so } C = -31.$$

27 Find C in the particular antiderivative $F(x) = x^3 + 5x + C$ of $f(x) = 3x^2 + 5$, satisfying $F(3) = 4$.

- a. 4
- b. -36
- c. -38 ✓
- d. -40

$$F(3) = (3)^3 + 5(3) + C = 42 + C = 4, \text{ so } C = -38.$$

28 Find C in the particular antiderivative $F(x) = -2x^3 + 7x + C$ of $f(x) = -6x^2 + 7$, satisfying $F(3) = -8$.

- a. 27
- b. 25 ✓
- c. -8
- d. 23

$$F(3) = -2(3)^3 + 7(3) + C = -33 + C = -8, \text{ so } C = 25.$$

29 Find C in the particular antiderivative $F(x) = x^3 + x + C$ of $f(x) = 3x^2 + 1$, satisfying $F(-1) = -7$.

- a. -7
- b. -3
- c. -4
- d. -5 ✓

$$F(-1) = (-1)^3 + (-1) + C = -2 + C = -7, \text{ so } C = -5.$$

30 Find C in the particular antiderivative $F(x) = -2x^3 + 6x + C$ of $f(x) = -6x^2 + 6$, satisfying $F(-3) = -8$.

- a. -8
- b. -46
- c. -44 ✓
- d. -42

$$F(-3) = -2(-3)^3 + 6(-3) + C = 36 + C = -8, \text{ so } C = -44.$$

31 Find C in the particular antiderivative $F(x) = 2x^3 + 8x + C$ of $f(x) = 6x^2 + 8$, satisfying $F(0) = 5$.

a. 7

b. 5 ✓

c. 3

d. 6

$$F(0) = 2(0)^3 + 8(0) + C = 0 + C = 5, \text{ so } C = 5.$$

32 Find C in the particular antiderivative $F(x) = -x^3 + C$ of $f(x) = -3x^2$, satisfying $F(2) = -1$.

a. 9

b. 5

c. -1

d. 7 ✓

$$F(2) = -(2)^3 + C = -8 + C = -1, \text{ so } C = 7.$$

33 Find C in the particular antiderivative $F(x) = 2x^3 + 4x + C$ of $f(x) = 6x^2 + 4$, satisfying $F(1) = 2$.

a. -2

b. -6

c. 2

d. -4 ✓

$$F(1) = 2(1)^3 + 4(1) + C = 6 + C = 2, \text{ so } C = -4.$$

Multiple choice — medium 15

34 Approximate $\int_0^4 5x^2 dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.

a. 150 ✓

b. 145

c. 170

d. 70

$\Delta x = 1$; right endpoints $x = 1, 2, \dots, 4$: $R_4 = 5[1^2 + 2^2 + \dots + 4^2] = 5(30) = 150$.

35 Approximate $\int_0^4 6x^2 dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.

a. 174

b. 84

c. 180 ✓

d. 204

$\Delta x = 1$; right endpoints $x = 1, 2, \dots, 4$: $R_4 = 6[1^2 + 2^2 + \dots + 4^2] = 6(30) = 180$.

36 Approximate $\int_0^6 2x^2 dx$ using a Right Riemann Sum with $n = 6$ equal subintervals.

a. 194

b. 110

c. 180

d. 182 ✓

$\Delta x = 1$; right endpoints $x = 1, 2, \dots, 6$: $R_6 = 2[1^2 + 2^2 + \dots + 6^2] = 2(91) = 182$.

37 Approximate $\int_0^7 6x^2 dx$ using a Right Riemann Sum with $n = 7$ equal subintervals.

a. 840 ✓

b. 546

c. 882

d. 834

$\Delta x = 1$; right endpoints $x = 1, 2, \dots, 7$: $R_7 = 6[1^2 + 2^2 + \dots + 7^2] = 6(140) = 840$.

38 Approximate $\int_0^3 5x^2 dx$ using a Right Riemann Sum with $n = 3$ equal subintervals.

a. 70 ✓

b. 85

c. 65

d. 25

$\Delta x = 1$; right endpoints $x = 1, 2, \dots, 3$: $R_3 = 5[1^2 + 2^2 + \dots + 3^2] = 5(14) = 70$.

39 Approximate $\int_0^3 2x^2 dx$ using a Right Riemann Sum with $n = 3$ equal subintervals.

a. 28 ✓

b. 34

c. 10

d. 26

$\Delta x = 1$; right endpoints $x = 1, 2, \dots, 3$: $R_3 = 2[1^2 + 2^2 + \dots + 3^2] = 2(14) = 28$.

40 Approximate $\int_0^4 4x^2 dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.

a. 120 ✓

b. 116

c. 136

d. 56

$\Delta x = 1$; right endpoints $x = 1, 2, \dots, 4$: $R_4 = 4[1^2 + 2^2 + \dots + 4^2] = 4(30) = 120$.

41 Approximate $\int_0^4 3x^2 dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.

- a. 42
- b. 90 ✓
- c. 102
- d. 87

$\Delta x = 1$; right endpoints $x = 1, 2, \dots, 4$: $R_4 = 3[1^2 + 2^2 + \dots + 4^2] = 3(30) = 90$.

42 Approximate $\int_0^7 x^2 dx$ using a Right Riemann Sum with $n = 7$ equal subintervals.

- a. 91
- b. 140 ✓
- c. 147
- d. 139

$\Delta x = 1$; right endpoints $x = 1, 2, \dots, 7$: $R_7 = 1[1^2 + 2^2 + \dots + 7^2] = 1(140) = 140$.

43 Approximate $\int_0^5 5x^2 dx$ using a Right Riemann Sum with $n = 5$ equal subintervals.

- a. 270
- b. 275 ✓
- c. 150
- d. 300

$\Delta x = 1$; right endpoints $x = 1, 2, \dots, 5$: $R_5 = 5[1^2 + 2^2 + \dots + 5^2] = 5(55) = 275$.

44 Approximate $\int_0^7 5x^2 dx$ using a Right Riemann Sum with $n = 7$ equal subintervals.

- a. 455
- b. 700 ✓
- c. 735
- d. 695

$\Delta x = 1$; right endpoints $x = 1, 2, \dots, 7$: $R_7 = 5[1^2 + 2^2 + \dots + 7^2] = 5(140) = 700$.

45 Approximate $\int_0^6 x^2 dx$ using a Right Riemann Sum with $n = 6$ equal subintervals.

a. 55

b. 91 ✓

c. 90

d. 97

$\Delta x = 1$; right endpoints $x = 1, 2, \dots, 6$: $R_6 = 1[1^2 + 2^2 + \dots + 6^2] = 1(91) = 91$.

46 Approximate $\int_0^5 3x^2 dx$ using a Right Riemann Sum with $n = 5$ equal subintervals.

a. 165 ✓

b. 162

c. 90

d. 180

$\Delta x = 1$; right endpoints $x = 1, 2, \dots, 5$: $R_5 = 3[1^2 + 2^2 + \dots + 5^2] = 3(55) = 165$.

47 Approximate $\int_0^5 2x^2 dx$ using a Right Riemann Sum with $n = 5$ equal subintervals.

a. 110 ✓

b. 120

c. 60

d. 108

$\Delta x = 1$; right endpoints $x = 1, 2, \dots, 5$: $R_5 = 2[1^2 + 2^2 + \dots + 5^2] = 2(55) = 110$.

48 Approximate $\int_0^4 2x^2 dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.

- a. 68
- b. 58
- c. 60 ✓
- d. 28

$\Delta x = 1$; right endpoints $x = 1, 2, \dots, 4$: $R_4 = 2[1^2 + 2^2 + \dots + 4^2] = 2(30) = 60$.

Multiple choice — hard 15

49 Approximate $\int_0^6 f(x) dx$ using the Trapezoidal Rule (equal width 2), given $f(0) = 7$, $f(2) = 12$, $f(4) = 9$, $f(6) = 3$.

- a. 54
- b. 52 ✓
- c. 62
- d. 50

$$T = 2 \left[\frac{7+12}{2} + \frac{12+9}{2} + \frac{9+3}{2} \right] = 2 \left[\frac{7+3}{2} + 12 + 9 \right] = 2(5 + 12 + 9) = 52.$$

50 Approximate $\int_0^3 f(x) dx$ using the Trapezoidal Rule (equal width 1), given $f(0) = 7$, $f(1) = 15$, $f(2) = 3$, $f(3) = 9$.

- a. 25
- b. 27
- c. 26 ✓
- d. 34

$$T = 1 \left[\frac{7+15}{2} + \frac{15+3}{2} + \frac{3+9}{2} \right] = 1 \left[\frac{7+9}{2} + 15 + 3 \right] = 1(8 + 15 + 3) = 26.$$

- 51 Approximate $\int_0^3 f(x) dx$ using the Trapezoidal Rule (equal width 1), given $f(0) = 4$, $f(1) = 7$, $f(2) = 12$, $f(3) = 2$.

a. 22 ✓

b. 25

c. 23

d. 21

$$T = 1 \left[\frac{4+7}{2} + \frac{7+12}{2} + \frac{12+2}{2} \right] = 1 \left[\frac{4+2}{2} + 7 + 12 \right] = 1(3 + 7 + 12) = 22.$$

- 52 Approximate $\int_0^6 f(x) dx$ using the Trapezoidal Rule (equal width 2), given $f(0) = 6$, $f(2) = 4$, $f(4) = 13$, $f(6) = 6$.

a. 44

b. 46 ✓

c. 48

d. 58

$$T = 2 \left[\frac{6+4}{2} + \frac{4+13}{2} + \frac{13+6}{2} \right] = 2 \left[\frac{6+6}{2} + 4 + 13 \right] = 2(6 + 4 + 13) = 46.$$

- 53 Approximate $\int_0^3 f(x) dx$ using the Trapezoidal Rule (equal width 1), given $f(0) = 1$, $f(1) = 5$, $f(2) = 11$, $f(3) = 9$.

a. 20

b. 21 ✓

c. 22

d. 26

$$T = 1 \left[\frac{1+5}{2} + \frac{5+11}{2} + \frac{11+9}{2} \right] = 1 \left[\frac{1+9}{2} + 5 + 11 \right] = 1(5 + 5 + 11) = 21.$$

- 54 For $f(x) = 4x^2$ on $[0, 5]$, determine whether the Left and Right Riemann Sums overestimate or underestimate $\int_0^5 f(x) dx$.

- a. Both Left and Right Riemann Sums underestimate.
- b. Both Left and Right Riemann Sums overestimate.
- c. Left Riemann Sum overestimates, and Right Riemann Sum underestimates.
- d. **Left Riemann Sum underestimates, and Right Riemann Sum overestimates. ✓**

$f(x) = 4x^2$ is increasing on $[0, 5]$, so left endpoints give smaller values (underestimate) and right endpoints give larger values (overestimate). So: Left Riemann Sum underestimates, and Right Riemann Sum overestimates.

- 55 For $f(x) = -x^2 + 36$ on $[0, 6]$, determine whether the Left and Right Riemann Sums overestimate or underestimate $\int_0^6 f(x) dx$.

- a. **Left Riemann Sum overestimates, and Right Riemann Sum underestimates. ✓**
- b. Left Riemann Sum underestimates, and Right Riemann Sum overestimates.
- c. Both Left and Right Riemann Sums overestimate.
- d. Both Left and Right Riemann Sums underestimate.

$f(x) = -x^2 + 36$ is decreasing on $[0, 6]$, so left endpoints give larger values (overestimate) and right endpoints give smaller values (underestimate). So: Left Riemann Sum overestimates, and Right Riemann Sum underestimates.

- 56 For $f(x) = 3x^2$ on $[0, 3]$, determine whether the Left and Right Riemann Sums overestimate or underestimate $\int_0^3 f(x) dx$.

- a. Left Riemann Sum overestimates, and Right Riemann Sum underestimates.
- b. Both Left and Right Riemann Sums overestimate.
- c. **Left Riemann Sum underestimates, and Right Riemann Sum overestimates. ✓**
- d. Both Left and Right Riemann Sums underestimate.

$f(x) = 3x^2$ is increasing on $[0, 3]$, so left endpoints give smaller values (underestimate) and right endpoints give larger values (overestimate). So: Left Riemann Sum underestimates, and Right Riemann Sum overestimates.

- 57 For $f(x) = 2x^2$ on $[0, 4]$, determine whether the Left and Right Riemann Sums overestimate or underestimate $\int_0^4 f(x) dx$.

- a. Left Riemann Sum overestimates, and Right Riemann Sum underestimates.
- b. Both Left and Right Riemann Sums underestimate.
- c. Left Riemann Sum underestimates, and Right Riemann Sum overestimates. ✓
- d. Both Left and Right Riemann Sums overestimate.

$f(x) = 2x^2$ is increasing on $[0, 4]$, so left endpoints give smaller values (underestimate) and right endpoints give larger values (overestimate). So: Left Riemann Sum underestimates, and Right Riemann Sum overestimates.

- 58 For $f(x) = -2x^2 + 8$ on $[0, 2]$, determine whether the Left and Right Riemann Sums overestimate or underestimate $\int_0^2 f(x) dx$.

- a. Both Left and Right Riemann Sums underestimate.
- b. Both Left and Right Riemann Sums overestimate.
- c. Left Riemann Sum overestimates, and Right Riemann Sum underestimates. ✓
- d. Left Riemann Sum underestimates, and Right Riemann Sum overestimates.

$f(x) = -2x^2 + 8$ is decreasing on $[0, 2]$, so left endpoints give larger values (overestimate) and right endpoints give smaller values (underestimate). So: Left Riemann Sum overestimates, and Right Riemann Sum underestimates.

- 59 A factory's production rate is $r(t) = 80 + 26t$ units/day, where t is days since a process change. Find the total production $P(2)$ given $P(0) = 0$.

- a. 160
- b. 238
- c. 186
- d. 212 ✓

$P(t) = \int r(t) dt = 80t + 13t^2 + C$. Using $P(0) = 0$: $C = 0$. $P(2) = 80(2) + 13(2)^2 = 160 + 52 = 212$ units.

- 60 A factory's production rate is $r(t) = 187 + 10t$ units/day, where t is days since a process change. Find the total production $P(15)$ given $P(0) = 0$.

- a. 2805
- b. 3920
- c. 3930 ✓
- d. 3940

$$P(t) = \int r(t) dt = 187t + 5t^2 + C. \text{ Using } P(0) = 0: C = 0.$$
$$P(15) = 187(15) + 5(15)^2 = 2805 + 1125 = 3930 \text{ units.}$$

- 61 A factory's production rate is $r(t) = 108 + 30t$ units/day, where t is days since a process change. Find the total production $P(19)$ given $P(0) = 0$.

- a. 7437
- b. 7467 ✓
- c. 2052
- d. 7497

$$P(t) = \int r(t) dt = 108t + 15t^2 + C. \text{ Using } P(0) = 0: C = 0.$$
$$P(19) = 108(19) + 15(19)^2 = 2052 + 5415 = 7467 \text{ units.}$$

- 62 A factory's production rate is $r(t) = 86 + 18t$ units/day, where t is days since a process change. Find the total production $P(14)$ given $P(0) = 0$.

- a. 2968 ✓
- b. 1204
- c. 2950
- d. 2986

$$P(t) = \int r(t) dt = 86t + 9t^2 + C. \text{ Using } P(0) = 0: C = 0.$$
$$P(14) = 86(14) + 9(14)^2 = 1204 + 1764 = 2968 \text{ units.}$$

63 A factory's production rate is $r(t) = 184 + 25t$ units/day, where t is days since a process change. Find the total production $P(8)$ given $P(0) = 0$.

a. 1472

b. 2272 ✓

c. 2247

d. 2297

$$P(t) = \int r(t) dt = 184t + \frac{25}{2}t^2 + C. \text{ Using } P(0) = 0: C = 0.$$

$$P(8) = 184(8) + \frac{25}{2}(8)^2 = 1472 + 800 = 2272 \text{ units.}$$